MATHEMATICS

Analysis and approaches

Standard Level | Set A | Paper 2

Mock examination · 2025 series

Time allowed: 1 hour 30 minutes Maximum mark: 80

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator is required.

• This paper has Section A and Section B. Both sections are compulsory.

• Answer every question. Show all working and give sufficient mathematical reasoning.

• A clean Mathematics: analysis and approaches formula booklet may be used.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.

• Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.

• The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 5]

Four observations, listed in ascending order, are $a, b, 10, 10$. Their median is 8 and their mean is 7.

(a) Find b. [1]

(b) Find a. [2]

(c) Find the population standard deviation of these observations. [2]

SECTION A

Question 2 [Maximum mark: 5]

Let $f(x) = \ln(x + 2)$ for $x > -2$, and $g(x) = e^x$ for $x \in \mathbb{R}$.

(a) Find $(g \circ f)(x)$. [2]

(b) Solve $(g \circ f)(x) = (f \circ g)(x)$. [3]

SECTION A

Question 3 [Maximum mark: 5]

A measurement X follows a normal distribution with mean 230 and standard deviation 9, so $X \sim N(230, 9^2)$.

(a) Find $P(X > 240)$. [2]

(b) Find a such that $P(a < X < 230) = P(X > 240)$. [3]

SECTION A

Question 4 [Maximum mark: 5]

Let $f(x) = x^2 \ln(x-2) + 1$, for $x > 2$.

(a) Find $f\prime(3)$. [2]

(b)(i)) Find an equation of the normal to the graph at $x = 3$ in point-gradient form. [2]

(b)(ii)) Write this equation in the form $y = mx + c$. [1]

SECTION A

Question 5 [Maximum mark: 6]

In the expansion of $(2x + ax^2)^7$, where $a \in \mathbb{R}$, the coefficient of x^9 is 6048.

Find all possible values of a. [6]

SECTION A

Question 6 [Maximum mark: 7]

A minor sector OAB has radius r centimetres and angle θ radians, where $0 < \theta < \pi$. The area between its chord AB and its minor arc is $3.35\,\text{cm}^2$. The perimeter of the sector, including both radii, is 16 cm.

A sector OAB and its minor segment

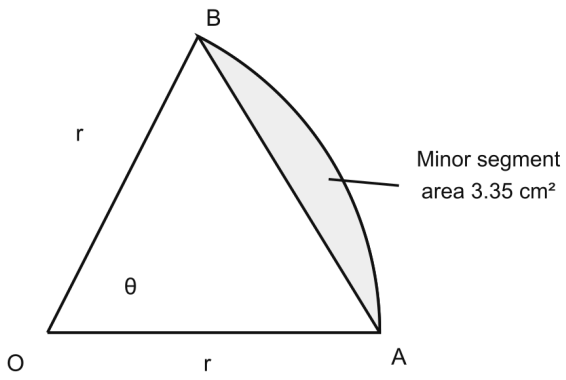


Diagram is schematic; θ is measured in radians.

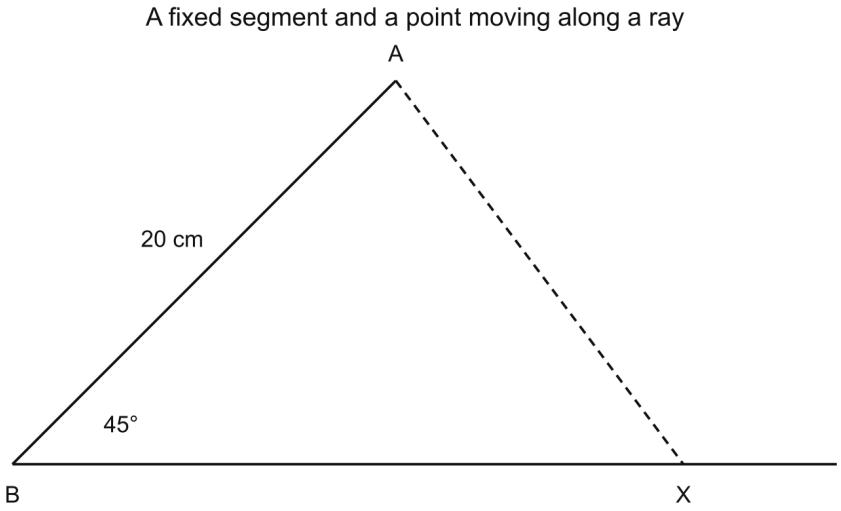
(a) Show that $r^2(\theta - \sin\theta) = 6.7$. [3]

(b) Find θ and r. [4]

SECTION A

Question 7 [Maximum mark: 7]

A fixed segment AB is 20 cm long. The point X lies strictly to the right of B on a horizontal ray. The angle ABX is $45°$. The diagram is schematic.



X varies along the ray to the right of B; not to scale.

(a) Find BX when AX is perpendicular to the ray. [2]

(b) Find BX when $AX = 20\text{cm}$. [1]

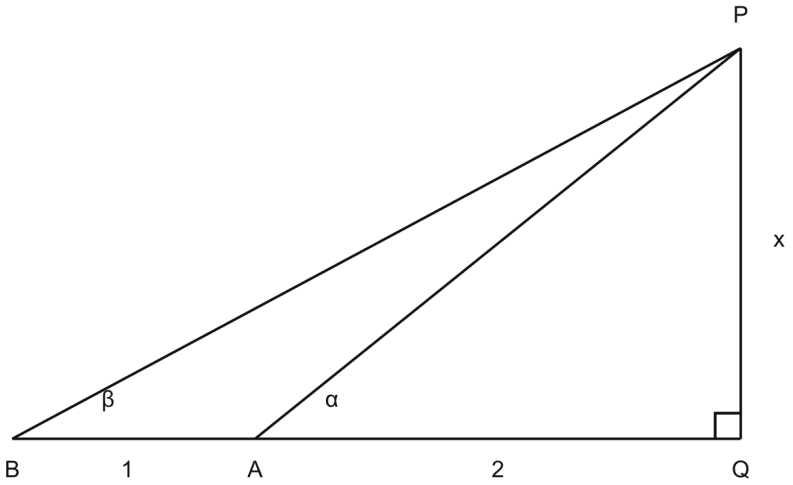
(c) Find the two possible values of BX when $AX = 17\text{cm}$. [4]

SECTION B

Question 8 [Maximum mark: 12]

Points B, A, Q lie in that order on a horizontal line, with $BA = 1$ and $AQ = 2$. Point P is vertically above Q, with $PQ = x > 0$. Let $\alpha = \angle PAQ$ and $\beta = \angle PBQ$. All angles are measured in degrees. Parts(a),(b),(c) describe separate positions of P.

Two angles of elevation from collinear points



(a)(i)) Given $\alpha = 45°$, find x. [1]

(a)(ii)) For this position, find β. [2]

(b) Given $\beta = 40°$, find α. [3]

(c)(i)) Now $\angle APB = 10°$. Show that $x = 2\tan\alpha = 3\tan\beta$. [2]

(c)(ii)) Express $\angle APB$ in terms of α and β. [1]

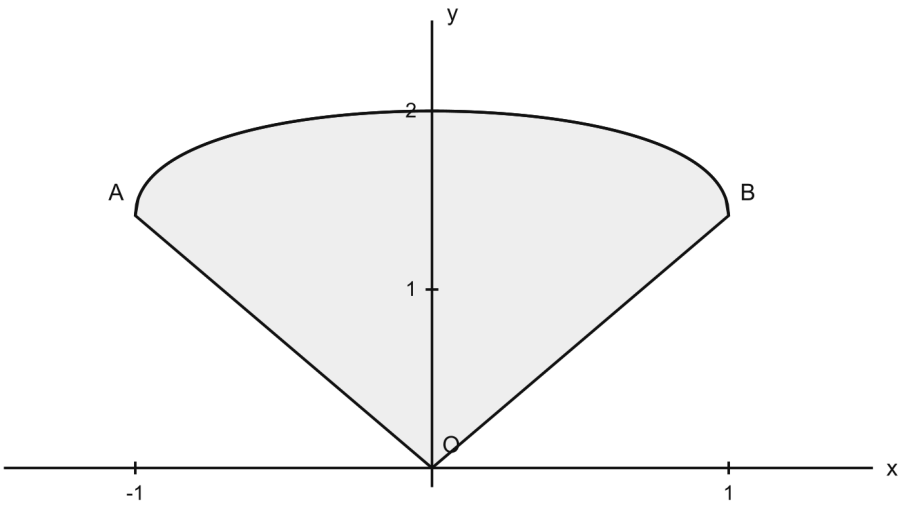
(c)(iii)) Find the two possible values of x. [3]

SECTION B

Question 9 [Maximum mark: 12]

Let $f(x) = \sqrt{1+x} + \sqrt{1-x}$, for $-1 \leq x \leq 1$. The endpoints of its graph are A and B, with B having positive x-coordinate. Let O be the origin.

Graph of f and the region bounded by the curve, OA and OB



(a)(i)) Find the exact value of $f(1)$. [1]

(a)(ii)) State the range of f. [2]

(b) The horizontal line $y = k$ meets the graph at exactly one point. Find k. [1]

(c) Find the gradient m of OB. [1]

(d) Find all values of k for which the line $y = kx$ has no point in common with the graph of f. [3]

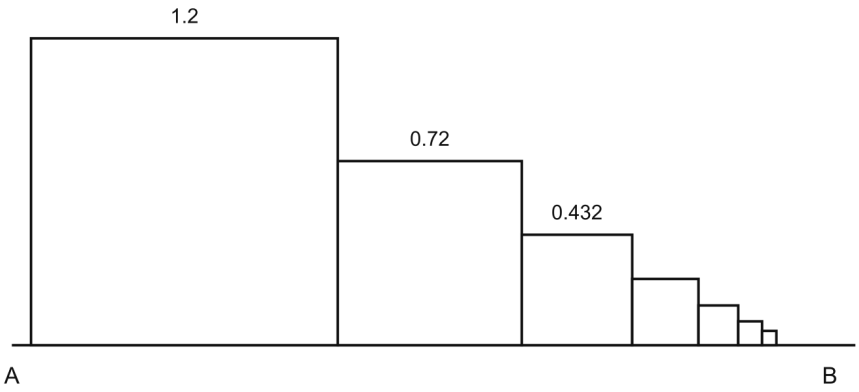
(e) Find the area of the shaded region bounded by the graph of f and the segments OA and OB. [4]

SECTION B

Question 10 [Maximum mark: 16]

An infinite chain of squares has consecutive bases on segment AB. The first three side lengths are 1.2,0.72 and 0.432 metres, and successive side lengths form a geometric sequence.

An infinite chain of adjacent squares



Squares continue indefinitely; bases lie consecutively on AB.

(a) Find the common ratio r of the side lengths. [2]

(b) Find the common ratio R of the areas. [2]

(c) Find the length AB. [3]

(d) Find the sum of the areas of the first 8 squares. Give your answer to 4 decimal places. [3]

(e) Find the sum of the areas of all the squares. [2]

(f) A separate chain has first side length $2x$ metres and side-length ratio x, where $0 < x < 1$. Its total base length is 8 m. Find the sum of its square areas. [4]

MATHEMATICS

Analysis and approaches

Standard Level | Set A | Paper 2

Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 5]

(a) The median is 8, hence $b = 6$. **A1**

[1 mark]

(b) $\frac{a+6+10+10}{4} = 7 \Leftrightarrow a+26 = 28 \Leftrightarrow a = 2$ **M1 A1**

[2 marks]

(c) . $\sigma \cong 3.32$ **A2**

[2 marks]

2. [Maximum mark: 5]

$f(x) = \ln(x+2)$, $g(x) = e^x$.

(a) $(g \circ f)(x) = e^{\ln(x+2)} = x+2$. **A1 A1**

[2 marks]

(b) $(g \circ f)(x) = (f \circ g)(x)$

$\ln(e^x + 2) = x + 2$ **A1**

$x \cong -1.16$ **A2**

[3 marks]

Remark

(b) can be solved analytically,

$\ln(e^x + 2) = x + 2$

$e^x + 2 = e^{x+2}$

$e^x + 2 = e^x e^2$

$e^x(1 - e^2) = -2$

$e^x = \dfrac{-2}{1-e^2}$

$x = \ln\left(\dfrac{-2}{1-e^2}\right) \quad (\cong -1.16)$

but it is not necessary!

3. [Maximum mark: 5]

(a) $P(X > 240) = 0.13326... \cong 0.133$. **M1 A1**

[2 marks]

(b) $P(X > a) = 0.5 + 0.13326... = 0.63326...$ **M1 A1**

Hence $a = 226.9355 \cong 227$ **A1**

[3 marks]

4. [Maximum mark: 5]

Let $f(x) = x^2 \ln(x-2) + 1$

(a) By GDC,

$f'(3) = 9$ **A2**

[2 marks]

(b) By GDC,

(i) $y - 1 = -\frac{1}{9}(x-3)$; **A1 A1**

(ii) $y = -0.111x + 1.33$ **OR** $y = -\frac{1}{9}x + \frac{4}{3}$ **A1**

[3 marks]

Remark

GDC gives all the results.

You can find the result analytically, but it will be more time consuming and risky!

For $x = 3$, $y = 9\ln 1 + 1 = 1$, so the point is (3,1)

$$f'(x) = 2x\ln(x-2) + \frac{x^2}{x-2}$$

$$f'(3) = 6\ln 1 + \frac{9}{1} = 9$$

and the results follow!

5. [Maximum mark: 6]

$^7C_r\left(2x\right)^{7-r}\left(ax^2\right)^r$ **M1**

The power of x is $7-r+2r=7+r$

$7+r=9 \Leftrightarrow r=2$ **A1**

Hence, the coefficient is $^7C_2\left(2x\right)^5\left(a\right)^2=672a^2$ **A1**

$672a^2=6048$ **M1**

$a^2=9 \Leftrightarrow a=\pm 3$ **A1 A1**

[6 marks]

6. [Maximum mark: 7]

(a) Area of sector – area of triangle **M1**

$\frac{1}{2}r^2(\theta-\sin\theta)=3.35 \Leftrightarrow r^2(\theta-\sin\theta)=6.7$ **A1 A1**

3 marks]

(b) $r\theta+2r=16 \Leftrightarrow r(\theta+2)=16 \Leftrightarrow r=\frac{16}{\theta+2}$ **A1**

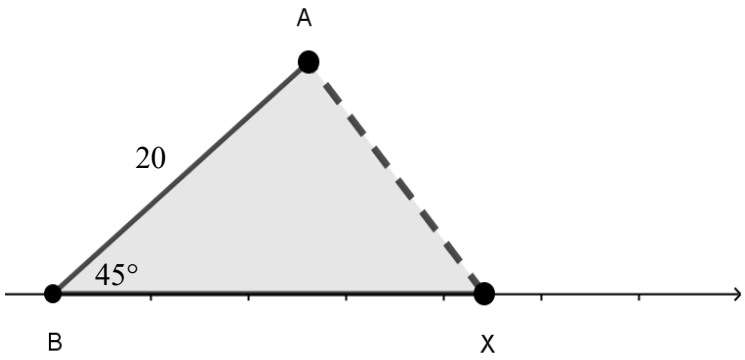
$\left(\frac{16}{\theta+2}\right)^2(\theta-\sin\theta)=6.7$ **M1**

$\theta=1.2$ **A1**

$r=5$ **A1**

[4 marks]

7. [Maximum mark: 7]



(a) $\cos 45° = \frac{BX}{20} \Rightarrow BX = 10\sqrt{2}$ **M1 A1**

[2 marks]

(b) $BX = 2 \times 10\sqrt{2} = 20\sqrt{2}$ **A1**

[1 mark]

(c) $17^2 = BX^2 + 20^2 - 2 \times 20 \times BX \cos 45°$. **M1 A1**

$BX = \sqrt{89} + 10\sqrt{2} \cong 23.6$ **A1**

$BX = -\sqrt{89} + 10\sqrt{2} \cong 4.71$ **A1**

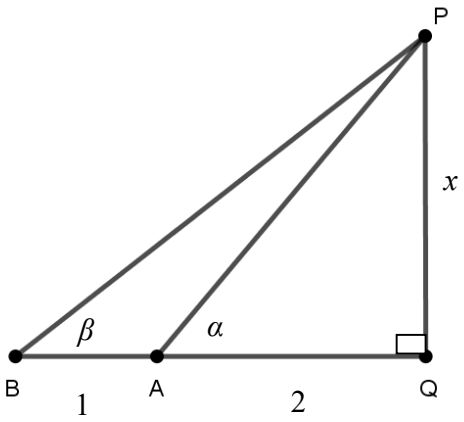
[4 marks]

Remark

For 7(c), you can also use sine rule to find the two triangles in the ambiguous case, although it is more time consuming.

Section B

8. [Maximum mark: 12]



(a) Given that $\alpha = 45°$

(i) $x = 2$. **A1**

(ii) $\tan\beta = \frac{2}{3} \Rightarrow \beta = 33.7°$. **M1 A1**

[3 marks]

(b) Given that $\beta = 40°$, find the size of α.

$\tan 40° = \frac{x}{3} \Rightarrow x = 2.51729....$ **M1 A1**

$\tan\alpha = \frac{2.51729...}{2} \Rightarrow \alpha = 51.5°$ **A1**

[3 marks]

(c) Let now $A\hat{P}B = 10°$.

(i) $\tan\alpha = \frac{x}{2} \Rightarrow x = 2\tan\alpha$ and $\tan\beta = \frac{x}{3} \Rightarrow x = 3\tan\beta$ **M1 A1**

Hence $x = 2\tan a = 3\tan\beta$.

(ii) $A\hat{P}B = 180° - \beta - B\hat{A}P = 180° - \beta - (180° - a) = \alpha - \beta$ **A1**

(iii) $\alpha - \beta = 10°$ $\alpha - \beta = 10° \Rightarrow \alpha = \beta + 10°$

$2\tan(10° + \beta) = 3\tan\beta$ **M1**

$\beta = 25.127...$ or $\beta = 54.872...$ **A1**

Hence, since $x = 3\tan\beta$

$x \cong 1.41$ or $x \cong 4.26$ **A1**

[6 marks]

9. [Maximum mark: 12]

$f(x) = \sqrt{1+x} + \sqrt{1-x}$, $-1 \le x \le 1$.

(a) (i) $\sqrt{2}$. **A1**

(ii) $\sqrt{2} \le y \le 2$ **A1 A1**

[3 marks]

(b) $k = 2$ **A1**

[1 mark]

(c) $m = \sqrt{2}$

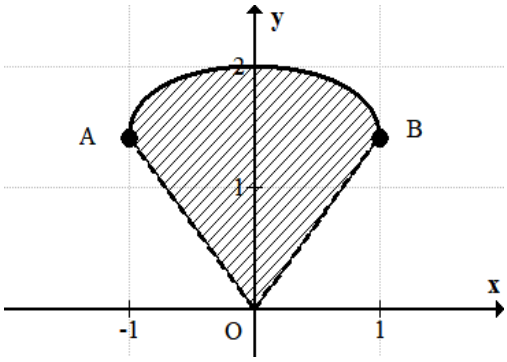
[1 mark]

(d) $-\sqrt{2} < k < \sqrt{2}$ **A1 A1**

(strict inequality) **A1**

[3 marks]

(e)



The area between the curve and x-axis is

$$\int_{-1}^{1} \left(\sqrt{1+x} + \sqrt{1-x}\right) dx = 3.77123... \quad \textbf{M1 A1}$$

Each triangle under the curve has area is $\frac{1 \times \sqrt{2}}{2} = \frac{\sqrt{2}}{2}$ **A1**

Hence, the area of the region is $3.77123... - \sqrt{2} \cong 2.36$ **A1**

[4 marks]

Remark

In (e), the exact value can also be found. It is $\frac{5\sqrt{2}}{3}$. But it is not necessary)

10. [Maximum mark: 16]

(a) $\frac{0.72}{1.2} = 0.6$, $\frac{0.432}{0.72} = 0.6$, $r = 0.6$ **M1 A1**

[2 marks]

(b) $\frac{0.72^2}{1.2^2} = 0.36$, $\frac{0.432^2}{0.72^2} = 0.36$, $R = r^2 = 0.36$ **M1 A1**

[2 marks]

(c) AB $= S_\infty$ for sides **M1**

AB $= \frac{1.2}{1-0.6} = 3$ **A1 A1**

[3 marks]

(d) For areas

$S_8 = \frac{1.2^2(1-0.36^8)}{1-0.36} \cong 2.2494$ **M1 A1 A1**

[3 marks]

(e) $S_\infty = \frac{1.2^2}{1-0.36} \cong 2.25$ **M1 A1**

[2 marks]

(f) For sides: $S_\infty = 8 \Leftrightarrow \frac{2x}{1-x} = 8 \Leftrightarrow x = 0.8$ **M1 A1**

For areas: $u_1 = (2 \times 0.8)^2 = 2.56$, $r = 0.8^2 = 0.64$ **A1**

$S_\infty = \frac{2.56}{1-0.64} = \frac{64}{9} \quad (\cong 7.11...)$ **A1**

[4 marks]